



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name:KAVIYA.G

Reg.No.:192524281

Course Code: CSA1709

Slot:B

Course Name: Artificial Intelligence

Course Faculty: Dr.P.Rajaram

Project Title: AI-Based Constraint Satisfaction Solver for Automated Timetable Scheduling

Module Photographs:

The composite image consists of three parts:

- Left:** A photograph of a student, Kaviya.G, standing next to a presentation screen. The screen displays the project title and a Google Maps location for Mevalurkuppam, Tamil Nadu, India. A laptop is on a table in front of her.
- Middle:** A detailed system architecture diagram titled "Module 3: Timetable Generation & Conflict Reporting - Architecture". It shows the flow between a Frontend (Web Interface), Backend (Timetable & Conflict Service Layer), External Modules, Data Storage, and Infrastructure & Cross-Cutting Concerns.
- Right:** A screenshot of the "AI Scheduler" web application interface. It features a sidebar with navigation options like Dashboard, Data Management, Conflicts, Console, and Reports. The main area displays a timetable table with columns for Day, Time, Subject, Faculty, Room, and Seats.

Project Description:

This module is responsible for displaying, analyzing, and reporting the timetable generated by the CSP solving engine through an interactive web interface. It presents the generated schedule with details such as day, time slot, subject, faculty, room, and student strength in an easy-to-understand timetable format. The module also evaluates the generated timetable for possible faculty conflicts, room conflicts, room capacity violations, and other constraint violations. It provides performance information such as constraint satisfaction rate, generation time, number of conflicts, and solution quality, and allows users to download the generated timetable as a CSV file for further use.

Kaviya.G

Student Signature

Guide Signature